



The Wrong Direction

Quick facilitation notes & answer key

Goal

Students fix a wrong-DIRECTION bug in one of two parallel scripts (C3.2 alter + C3.1 write). The goal is 'both reach 2'; Bit is correct, but Pixel moves forward (away from 2, overshooting to 5) when it should move back. Students reason about where the goal is relative to the sprite's home and change the direction. This builds direction sense alongside debugging.

Ontario curriculum (Grade 2 — C3 Coding)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential and concurrent events.

C3.2 — read and alter existing code, including code that involves sequential and concurrent events, and describe how changes to the code affect the outcomes.

Materials

The printed worksheet and a pencil. Optional: a star on square 2; place a counter on Pixel's home (4) and ask 'is the star in front or behind?' before choosing the direction.

Run it (about 20 min)

1. I DO: state the goal — 'both reach 2.' Run Bit: 0 forward 2 → 2 (correct).
2. WE DO: run Pixel: 4 forward 2 → tries 6 but stops at 5 (the path ends at 5). 'Pixel went the wrong way — away from 2.'
3. YOU DO: students see the goal 2 is BEHIND Pixel (which is at 4), so the move should be back 2; they fix it and confirm Pixel reaches 2.
4. Connect: 'Check the direction: if the goal is behind the sprite, the move must go back.'

Answer key (modelled by the run-gate)

Ex 1 — Bit: 0 → forward 2 → 2 (reaches goal). Pixel: 4 → forward 2 → 5 (the path stops at 5; Pixel overshoots and misses 2).

Ex 2 — the goal 2 is BEHIND Pixel (Pixel is at 4), so Pixel should move BACK, not forward.

Ex 3 — fixed move: back 2 → Pixel lands on 2. Now BOTH reach the goal 2.

Fix note: the number (2) was fine; only the direction changes from forward to back.

Watch for & differentiate

Common error: changing the number instead of the direction — here the count of 2 is correct; the direction is the bug.

Common error: forgetting the path stops at 5 — forward from 4 cannot reach 6, so Pixel sits at 5.

Simplify: ask 'is the star in front of Pixel or behind it?' to choose the direction.

Extend: ask which direction Bit would need if Bit started at 4 like Pixel (back 2 as well).

Success indicator: the child changes Pixel's move to back 2 and confirms both reach 2.